

Tyler Morrill

Software Engineer & Research Scientist

Los Angeles, CA
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Experience

December 2022 - Present

Software Engineer II — *Lockheed Martin* (Los Angeles, CA)

- Core developer on a team writing a parallel, high-performance physics simulation software suite for use on GPUs and supercomputers.
- Research and implement numerical solver techniques such as iterative solvers, matrix compression, preconditioning, and interpolation.
- Write codes, tools, and other scripts to support engineers in iterative design workflows.
- Provide support and assistance to code users, write/maintain code documentation, create software tutorials, and present technical material to users and stakeholders.

January 2020 - May 2022

Teaching Assistant & Lead Math Tutor — *UCF* (Orlando, FL)

- Assist course instructor during lectures for Calculus and Differential Equations courses.
- Hold recitation sessions and office hours; grade papers.
- Provide both one-on-one and group assistance to students in courses ranging from college algebra to differential equations.

Education

January 2026 - Present

Georgia Institute of Technology

Computer Science M.S. – Computer Graphics

January 2020 - May 2022

University of Central Florida

Mathematics B.S. & Computer Science Minor – Computational Mathematics

Projects

Real-time Fluid Simulation — [Github](#)

- Real-time, interactive fluid simulation project using a variation of the Smoothed Particle Hydrodynamics (SPH) method.
- Rendered using a custom-built Vulkan renderer (see below).
- Using fast ODE solvers and unbounded spatial hashing.
- Run using multithreaded CPU impl. or on GPU with compute shaders.

Rendering Engine API — [Github](#)

- Low-abstraction, performance-oriented, extensible real-time rendering engine using Vulkan API to streamline its use in my own graphics projects.
- Support for shaders written in Slang (and thus GLSL or HLSL too).
- OBJ and GLTF model importing (geometry and scene hierarchies).
- Modern Vulkan features like dynamic rendering, buffer device addresses, and descriptor heaps.
- Windows and Linux compatibility.

Skills

C, C++, Python, Fortran, C#, Odin proficiency

Concurrent programming using OpenMP, MPI, and manual multithreading

Unity and Godot game development

Profiling with RenderDoc, gperftools, and other homebrew profilers

Vulkan and OpenGL graphics APIs

GPU shader programming with GLSL, HLSL, Slang

Adept in Linux development environments

Large Git repository management

MATLAB and Python visualization scripting

Technical presentation to technical and non-technical audiences

Writing documentation, tech memos, and seminars

Honors & Awards

Cum Laude Honors

Distinguished

Undergraduate Researcher

Award: UCF

Hernandez Award for Mathematics

Pi Mu Epsilon Mathematics Honor Society

Dean's List Academic

Honor Roll Pegasus

Scholarship: UCF